



University of Louisville
Computer Science and Engineering Department

DATA MINING – CSE 632

Fall 2025

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ASSIGNMENT: Project 1

In this project, you will practice data preprocessing by downloading raw data from data.gov website. This particular one pertains to crime incidents in the city of Los Angeles starting from 2020 and it can be found in the following link:

<https://catalog.data.gov/dataset/crime-data-from-2020-to-present>

The website contains information about the dataset as well as the dataset in various formats. Please download and use the csv file format. Also, download and review the metadata file as well as check the original source of the data by checking the “landing page” of the source.

1. Explore the website and explore data mining used in crime data analysis

Explore the website and get used to the various files of dataset as well as metadata and the landing page. Find the descriptions of data features on the website as well.

Review three published academic papers on data mining applications in crime data analysis. Summarize these papers in your project report.

2. Explore Your Data

Understand the features in your data. Discuss the following aspects of the data in your project report:

1. What feature has missing data? How much data is missing? How do you deal with missing data?
2. What features are potentially noisy? What level of ambiguity is in the data? How do you deal with such noise?
3. What features need normalization? What feature might need discretization?
4. Are there any features that need to be created from unstructured data?
5. Are there any other preprocessing steps needed for the data?

3. Hypothesis of what might reflect crime trends

Once you are familiar with the structure and content of the data, **you need to state in your project three aspects of the data that you think might help you show certain crime trends.** You can choose to test for any crime trend you believe in. An example hypothesis is “White Collar crime like identity theft is getting less common while violent crime is on the rise.” You may use the example hypothesis and two of your own or come up with all three of your own hypotheses. You will visualize your data to prove/disprove your hypotheses.

Write one paragraph for each hypothesis of change you decided to explore. State your hypothesis and explain why you think this hypothesis is essential in each paragraph.

4. Preprocessing the Data

This is the most important part of the project. **You will perform at least five preprocessing techniques on the dataset.** Do not randomly choose any techniques to perform. Keep your end goal of visualizing your hypothesis in mind. Your preprocessing should help you to visualize and discuss your hypothesis later.

For each preprocessing technique, discuss the following in your project. Each point should have at least one paragraph of discussion:

1. Explain what technique is used and how it works.
2. Detailed explanation of how the technique is applied. If the technique requires parameters, explain them and how they are chosen.
3. Show and discuss the result of the technique. **Do not copy and paste raw results or screen capture.** Summarize your results into figures and explain what the figures say about your preprocessing results.
4. Explain how this technique helps you discuss your hypothesis later.

5. Data Visualization

After the data is preprocessed, you should try to have three different types of visualizations for your three hypotheses. You can use multiple visualizations for a single hypothesis if needed, but you should have at least have one for each hypothesis. You are looking for patterns that confirm or disprove your hypothesis. Patterns should be spatial, temporal, or demographic aspects of the data. Discuss your visualization and explain how it shows evidence of effects of climate change through time.

Any visualization tool, such as Excel or Tableau Public, is OK. The project is not graded on the tools used but on the discussion. Make sure you explain what each visualization shows in detail. Discuss the following aspects for each visualization:

1. Why is this visualization created?
2. How is it created?
3. What does each element of the visualization mean?
4. What pattern does this visualization show?
5. What does this pattern say about your hypothesis?
6. Is your hypothesis correct or not based on this visualization?

**Please read “Required Structure for the Project 1 Documentation” and
“Required Materials for Project 1 Submission” below.**

NOTE: Many software tools give a large number of output results and graphical interpretations (tens of pages of graphs, tables, diagrams!). Don't use them in your report if you don't understand these values and graphs, and if you don't have enough knowledge to make an interpretation and give appropriate discussion about these results. Just attachment to the project of these kinds of "software listings" without interpretation will have a negative influence on the final project grade.

Your interpretation and analysis should be explained in detail rather than just listing the techniques and their results.

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Required Structure for the Project 1 Report

1. *Cover page:* Course, Title of the project, student name, and date.
2. *Abstract:* Short description of the problem and the solution given in the project applied techniques and obtained results (one page max).
3. **Project: TITLE**
 - 3.1 **Problem description.** Explain crime statistics gathering and data mining in crime data analysis.
 - 3.2 **Data Gathering.** Explain how you gathered your data.
 - 3.3 **Data Exploration.** Explore your data. Discuss aspects of the dataset.
 - 3.4 **Hypothesis.** Based on your data exploration result, clearly state your three hypotheses.
 - 3.5 **Preprocessing.**

Software tools. Explain what tools you used for preprocessing.

At Least five preprocessing Techniques. Explain in detail what techniques are applied and the results.
 - 3.5 **Visualizing patterns:**

Software tools. Explain what tools you used for visualization

Visualizations. Explain in detail what each visualization means.

- 3.6 **Conclusion:** Do your findings reflect what news says or what your real-life experiences tell you? What can you learn from your findings?

Appendix:

Example citation format:

MLA format: Fayyad, Usama M. "Data mining and knowledge discovery: Making sense out of data." *IEEE Expert: Intelligent Systems and Their Applications* 11.5 (1996): 20-25.

APA format: Fayyad, U. M. (1996). Data mining and knowledge discovery: Making sense out of data. *IEEE Expert: Intelligent Systems and Their Applications*, 11(5), 20-25.

Harvard format: Fayyad, U.M., 1996. Data mining and knowledge discovery: Making sense out of data. *IEEE Expert: Intelligent Systems and Their Applications*, 11(5), pp.20-25.

Project 1 presentation

In this project you will also need to give a presentation about the entire process. The emphasis will be on whether the students properly explain their reasoning behind making certain decision. As the project is open-ended and there isn't a "only correct way", you will have to make choices based on your understanding and reasoning, and in the presentation, you will be explaining those. Simply explaining the graphs and repeating what is in the project report will not yeild a good grade. The presentation will contain 2 parts:

1. presentation slides
2. presentation video (length between 5 and 10 minutes)

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Required Materials for the Project 1 Submission

- 1. Project Report (80%)**
- 2. Project Presentation slides and video (5-10 mins long) (20%)**

Due date for Project 1:

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